

TABLE I
A Comparison: Select General-Purpose, Mid-Range Multimeters
in the Market Today (International Brands)

Brand	Model	Specifications											
		Range							Product warranty	Safety	Battery life	Weight	Price
Volt (AC)*	Volt (DC)	Resistance	Capacitance	Diode	Current (AC)	Current (DC)							
BK Precision	2709B	0.1mV - 750V	0.1mV - 1000V	0.1Ω - 66MΩ	1pF - 66mF	Up to 3.5V	0.1μA - 10A	0.1μA - 10A	Three years	IEC61010-1 (EN61010-1), CAT II 1000V, CAT III 600V, Class II	150 hrs	285g	US\$ 105**
BK Precision	2712	0.01mV - 750V	0.01mV - 1000V	0.01Ω - 40MΩ	1pF - 40μF	Up to 3V	1μA - 10A	1μA - 10A	Three years	IEC61010-1 (EN61010-1), CAT II 1000V, CATIII 600V, Class II	150 hrs	285g	US\$ 115**
Fluke	15B+	0.1mV - 1000V	0.1mV - 1000V	0.1Ω - 40MΩ	0.01nF - 1000μF	1mV - 2V	0.1μA - 10A	0.1μA - 10A	One year	IEC 61010-1, IEC61010-2-030 CAT III 600V, CAT II 1000V	500 hrs	455g	₹ 6888 **
Fluke	17B+	0.1mV - 1000V	0.1mV - 1000V	0.1Ω - 40MΩ	0.01nF - 1000μF	1mV - 2V	0.1μA - 10A	0.1μA - 10A	One year	IEC 61010-1, IEC61010-2-030 CAT III 600V, CAT II 1000V	500 hrs	455g	₹ 8483 **
GW Instek	GDM 360	0.01mV - 750V	0.01mV - 1000V	0.1Ω - 60MΩ	0.01nF - 4000μF	1mV - 2V	0.1μA - 10A	0.1μA - 10A	One year	EN 61010-1, EN 61010-2-030 CAT.III 1000V, CAT.IV 600V	N/A	370g	₹ 5700
Keysight	U1230	0.1mV - 600V	0.1mV - 600V	0.1Ω - 60MΩ	1nF - 10mF	1mV - 2V	0.01μA - 10A	0.01μA - 10A	Three years	EN/IEC 61010-1:2001, ANSI/UL 61010-1:2004, and CAN/CSA-C22.2 No. 61010-1-04, CAT III 600V	500 hrs	365g, 371g	₹ 7986 ***
Yokogawa	TY530	0.1mV - 1000V	0.1mV - 1000V	0.1Ω - 60MΩ	0.01nF - 1000μF	1mV - 2V	0.1μA - 10A	0.1μA - 10A	Three years	EN61010-1, EN61010-2-030, EN61010-031, 1000V CAT III, 600V CAT IV	300 hrs	570g	US\$ 248.99 - US\$ 270.68**

For accuracy and detailed specifications, please refer to the specifications of the particular multimeter at company's website

* True RMS

**Price as quoted on e-commerce websites

***Price as quoted on company's website



Fig. 2: Keithley's DMM7510

more intuitive, interesting and easy to learn for users. Claimed to be the first in a new class of digital multimeters, DMM7510 from Keithley Instruments is an integration of

1MS/sec, 18-bit digitiser for waveform capture, 12.7cm (5-inch) capacitive touch-screen display and, of course, a digital multimeter with resolution levels ranging from 3 1/2 to 7 1/2 digits.

The touch interface allows you to set up test faster and view results numerically as well as graphically on the screen with the ability to pan, pinch and zoom the waveform or signals, making high interaction with the data possible.

In a release, Jerry Janesch, senior market development manager at Keithley, noted that their DMM 7510 aids engineers to capture small signals at greater accuracy and pace in comparison to traditional digital multimeters currently in the market. Engineers get an in-depth understanding of their devices under test. To enhance low-level accuracy, this multimeter has expanded measurement ranges (100mV, 1Ω and 10μA). DMM7510 looks to address a wide range of test applications, including device characterisation, debugging and analysis, production or automated test, and